

Anomaly Detection in Traffic Data

Nahush Kolhe

Supervised by Prof. Edward Chung

The Hong Kong Polytechnic University

30th May 2024

A solid teal-colored horizontal bar spanning the entire width of the slide at the bottom.

Table of Contents

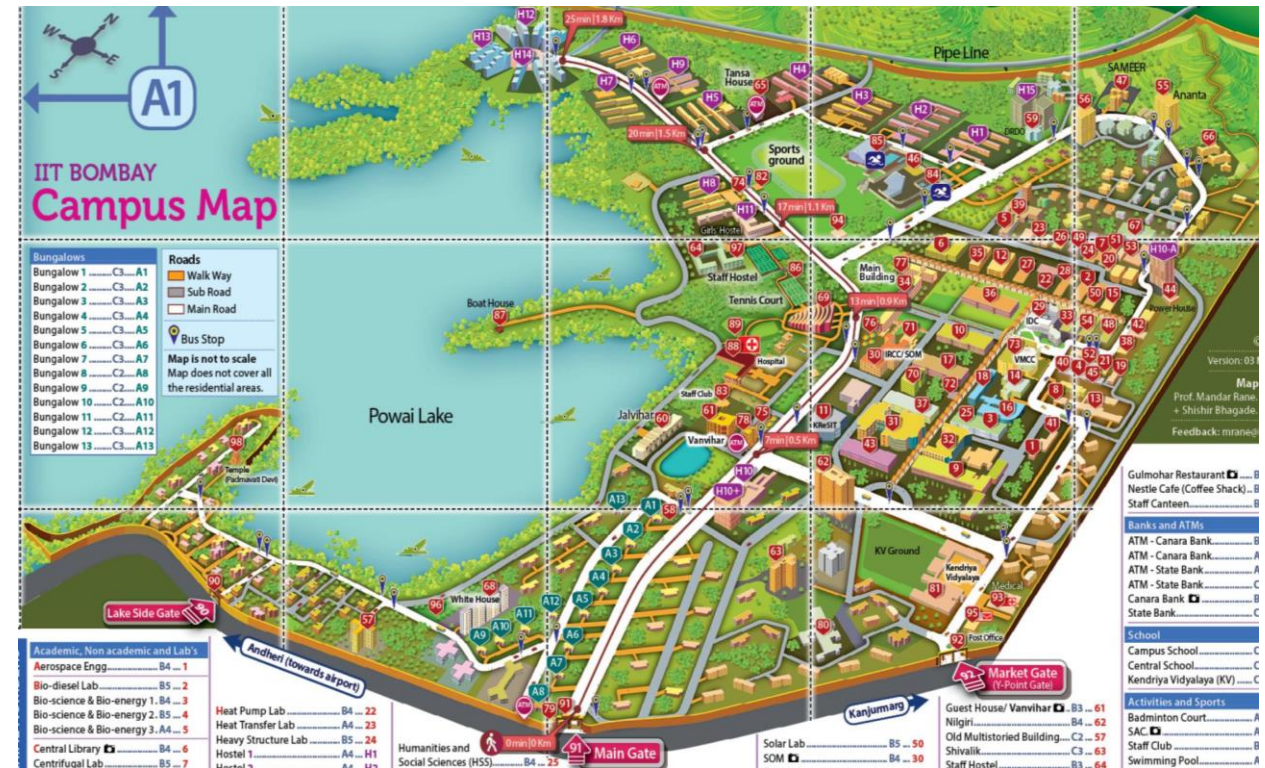
- About me & IIT Bombay
- Introduction
- Methodology
- Results
- Conclusion

About Me & IIT Bombay

- I am Nahush Kolhe, a 3rd year UG in Engineering Physics at IIT Bombay.
- Interest in Machine Learning, also pursuing minor in Data Science
- Indian Institute of Technology, Bombay (IITB), one of the best engineering college in India.
- Entrance exam : JEE Advanced, (2 yearlong preparation)
- Several branches : CS, EEE, Aerospace, Physics, Chemical, Metallurgy, etc.
- Good research opportunities to work under Professors
- Freedom to explore other branches & non-core as well like Finance, Consult etc.

Campus Life

- Huge (525 acre), and green campus.
- Hostels, Academic area, Residential area, Gymkhana grounds, Student Activity Area, and even a hill!
- Everything is student run, like clubs.
- Nightlife, especially at canteens.
- One of the largest cultural and technical fests in Asia, 'Mood Indigo' and 'Techfest'.



Map of IIT Bombay

Introduction

Introduction

- **Anomaly** : is a data point or pattern that significantly deviates from the expected behavior or norm in a dataset.
- **Role of Detector** : Cameras which capture images of traffic and processes it further.
- **Road Traffic Data** : Information about speed, flow, and occupancy of vehicles.

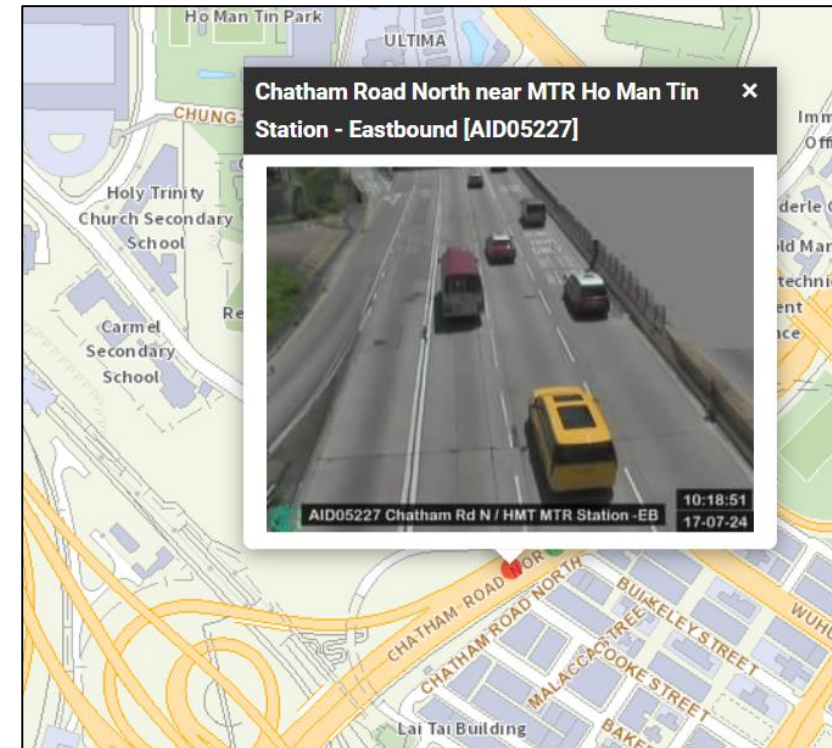


Image captured by detector

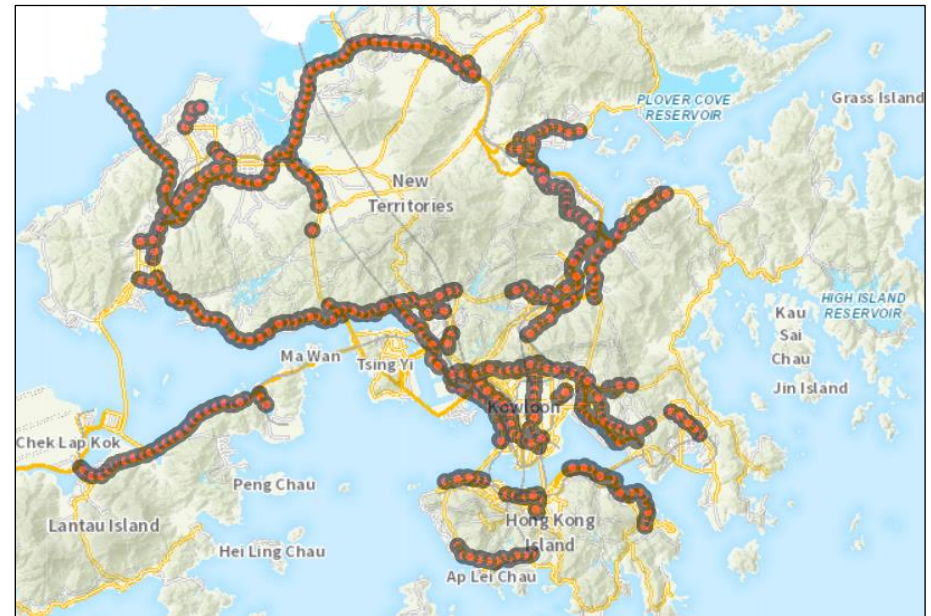
(available on www.hkemobility.gov.hk)

Problem Statement : Automatically detect anomalies in the traffic data for a network of detectors.

- We analyse detectors on strategic/major routes of Hong Kong

Why do this ?

1. Identify anomalies to improve data collection.
2. To obtain clean data without anomalies, which is used further by stakeholders.



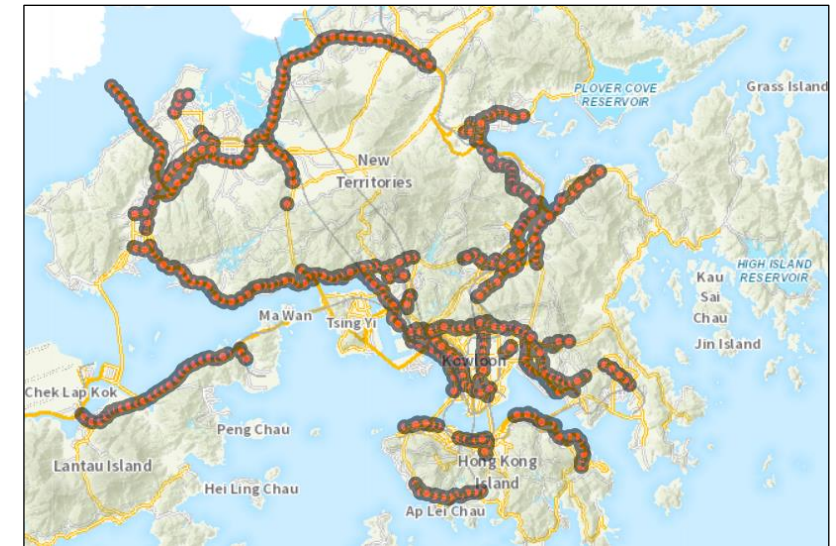
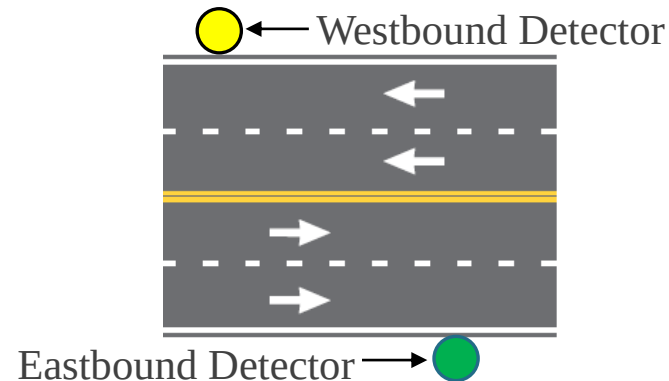
Detectors on Major routes of Hong Kong
(available on portal.csd.gov.hk)

Dataset Description

- Obtained from government website (data.gov.hk) of Hong Kong.
- 6 months data (10/23 - 03/24) for 768 detectors.

Data 1. Contains speed, flow & occupancy for each lane of detector sampled at 2-min frequency

Data 2. Contains Location of all detectors & their bound direction.



Location of detectors on major routes

Methodology

1. Correlation between nearby detectors : Identifies evident anomalies, and helps to create training datasets for further methods

2. Profile Model

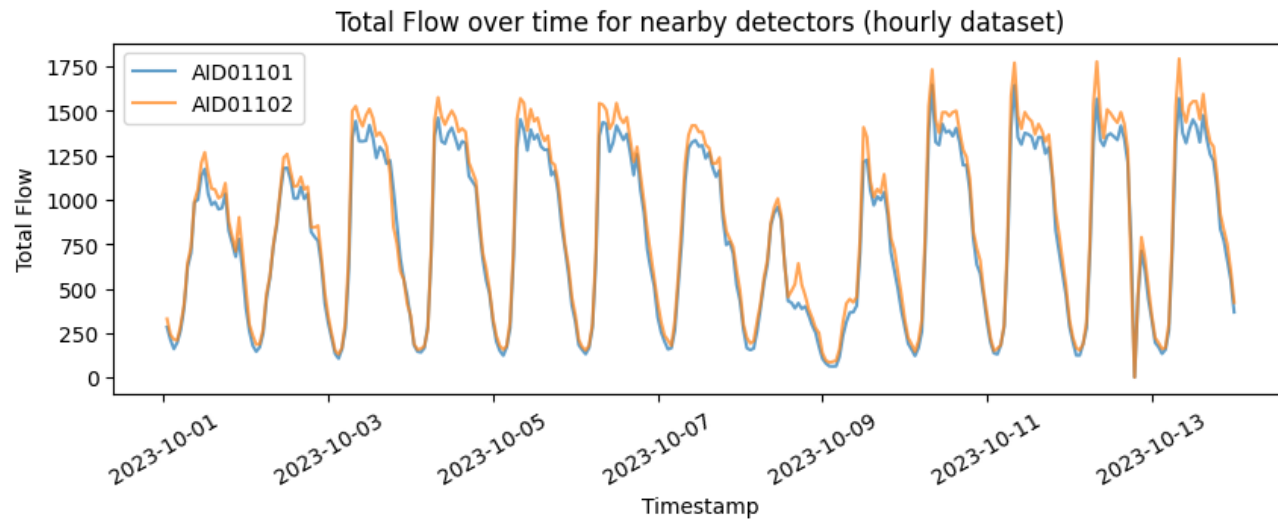
3. Supervised Classification



Machine Learning Models

1) Correlation between Nearby Detectors

- Nearby detectors along the same road should be highly correlated

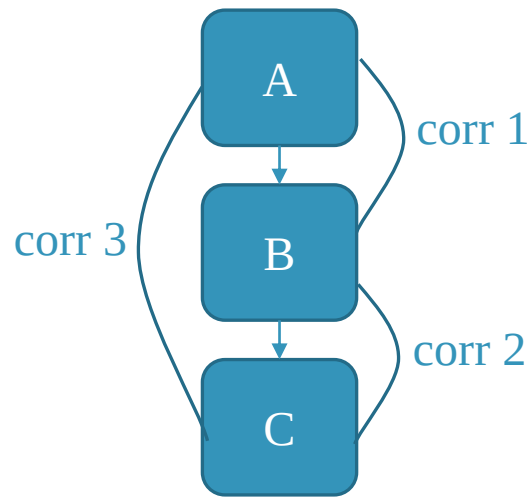


Example of nearby detectors being highly correlated (>0.9)

- Total flow is found to be the best metric to classify anomalies, due to conservation of flow.
- Using the location, we can obtain triplets of detectors which should be highly correlated.

Triplets of Detectors

- **Why triplets** : It gives just enough information to identify the data as anomaly or not.
- Exploiting the systematic naming of detectors and ensuring that they lie within 1km, we obtain triplets of detectors.
- Calculate the correlation within the triplets to identify good data (without anomalies)



Detectors A,B,C

Correlation value			Detectors		
corr 1	corr 2	corr 3	A	B	C
✓	✓		✓	✓	✓
✓	✗		✓	✓	✗
✗	✓		✗	✓	✓
✗	✗	✓	✓	✗	✓
✗	✗	✗	✗	✗	✗

Classification for a triplet (A, B, C)

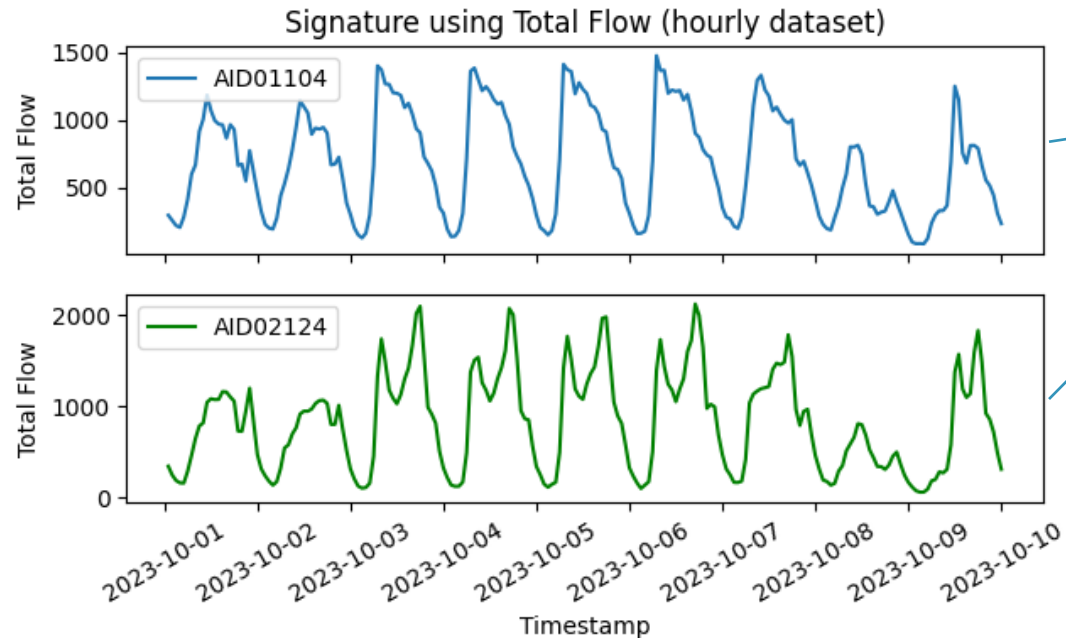
Threshold used is 0.9 for good data

corr > 0.9 → ✓

corr < 0.9 → ✗

2) Profile Model

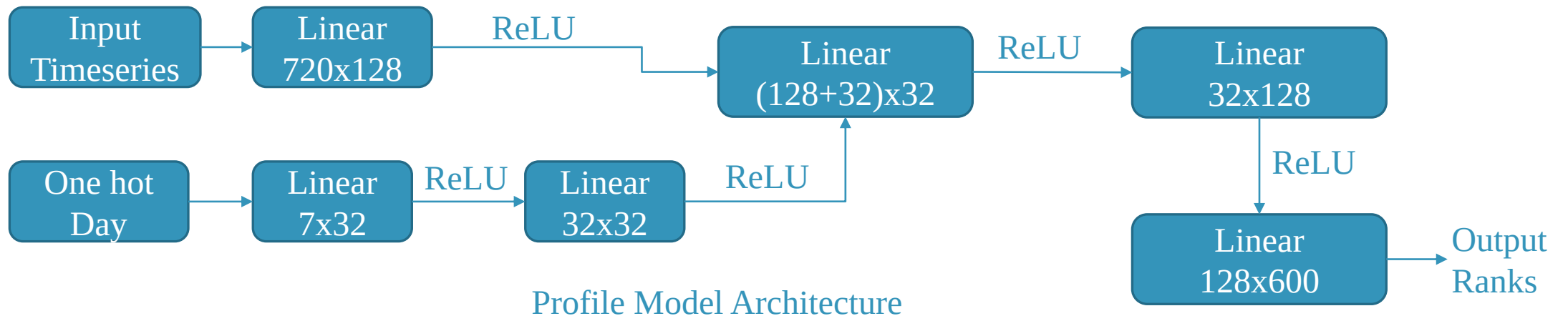
- It learns the signature of the detector.



- Different signature for different detectors.
- If the model doesn't identify the detector by its signature, then its an anomaly

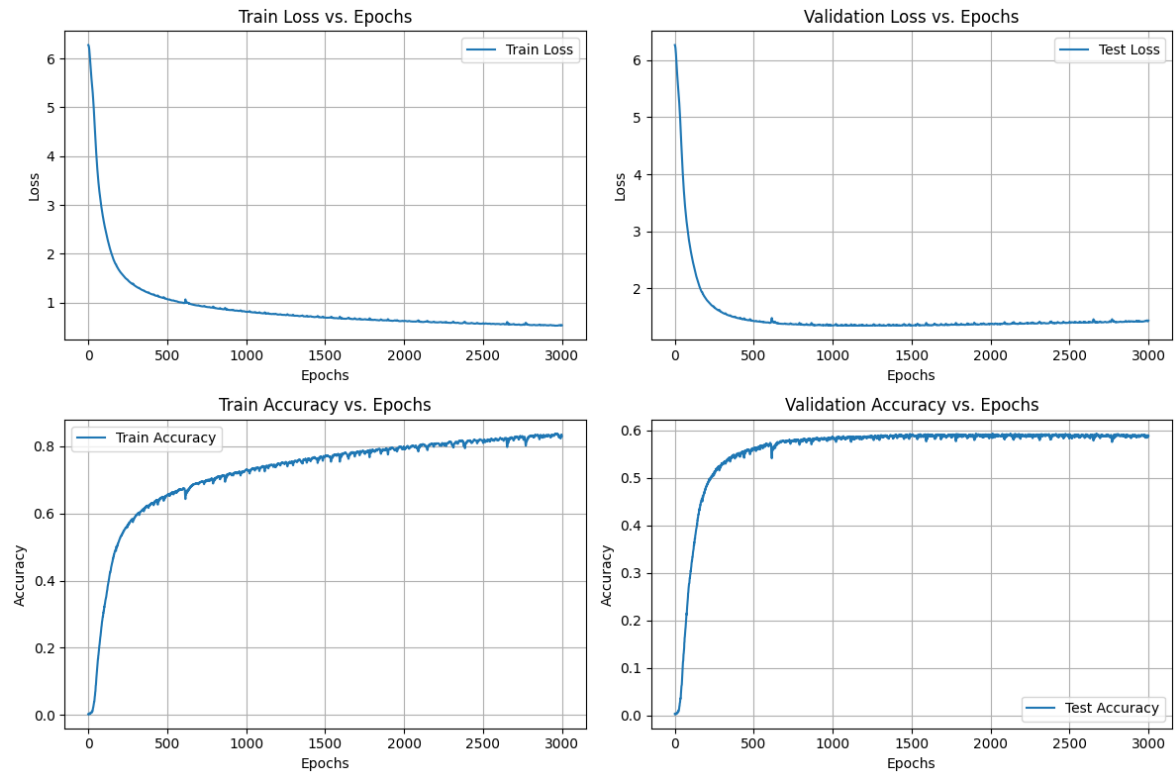
2) Profile Model

- Machine Learning Classification model (600 classes - for each detector)
- Input : 2-min timeseries flow data for a day and expect the model to identify the detector correctly.
- Output : Ranks of detectors (rank 1 = best predicted detector)



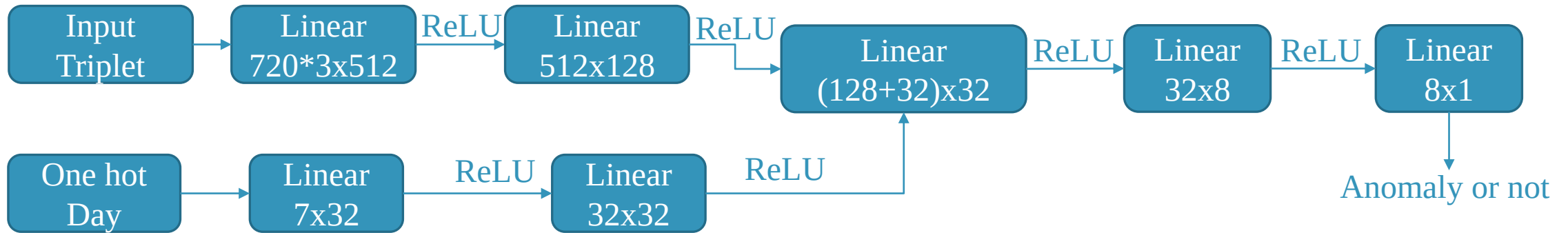
Training Profile Model

- We use the triplets to get only the highly correlated (>0.9) data for training.
- Remove data which has $>5\%$ null values.
- Profile Model with Cross-entropy loss.
- Optimizer = Adam, lr = 0.001, wd = 0.001
- **Top 1% Accuracy** : Accuracy that correct detector lies in top 1% of the predicted.
- Top 1% Accuracy for Test data = 92%



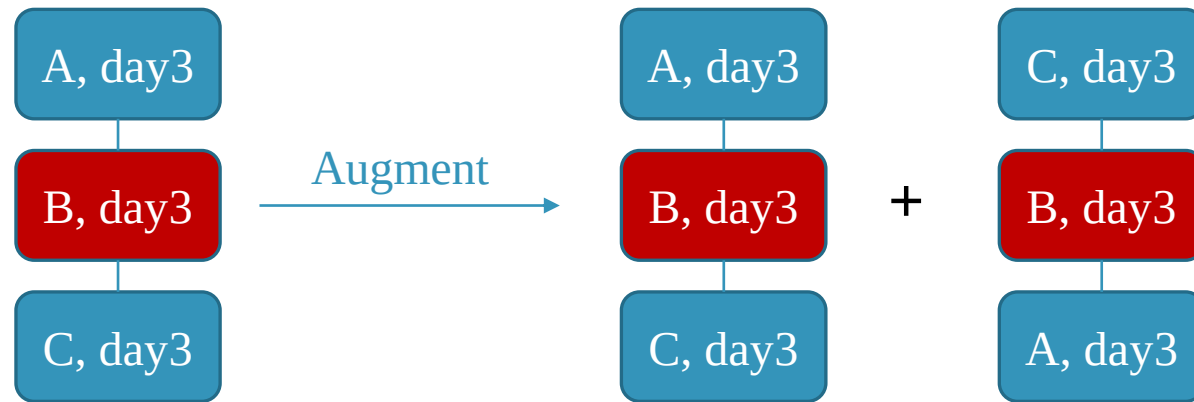
3) Supervised Classification

- We use triplet data to find the average correlation of a detector for each day.
- Label highly correlated (>0.9) as 1, and very low correlation (<0.7) as 0 (anomaly)
- Input for a detector on a day will also contain data from its triplets, and output is the label.



Training the Supervised Model

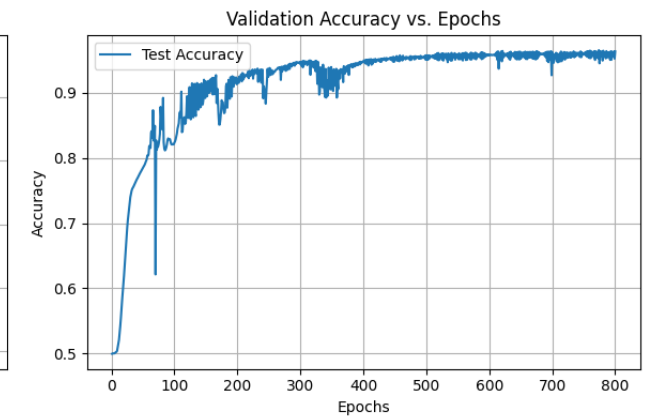
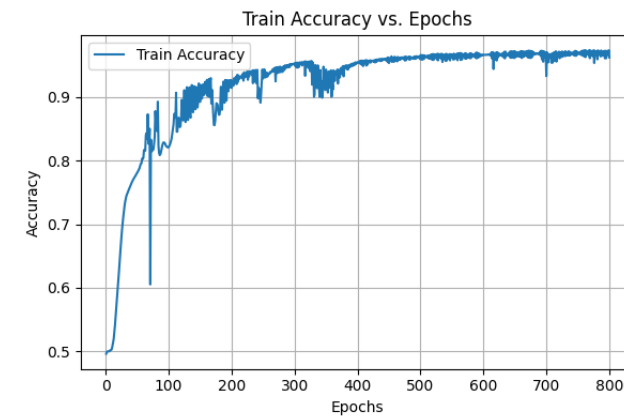
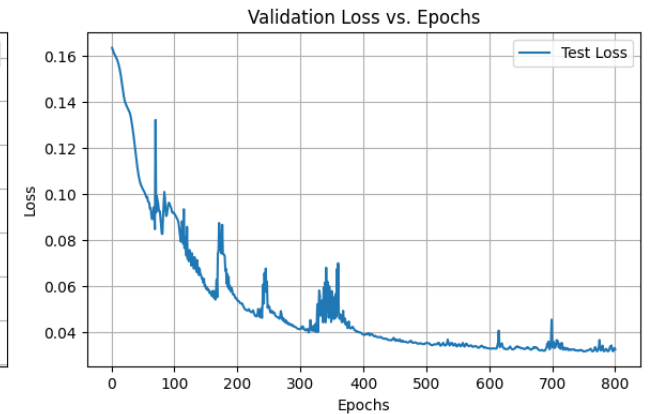
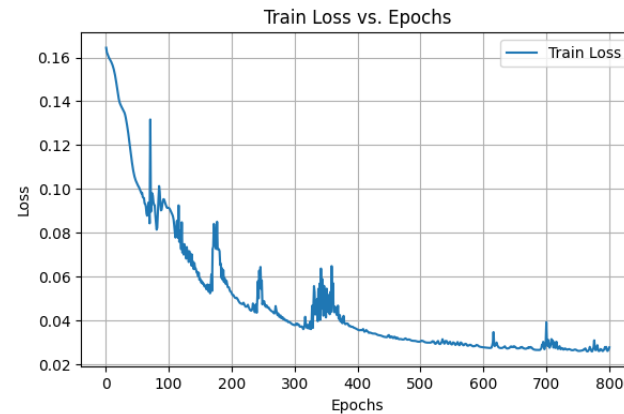
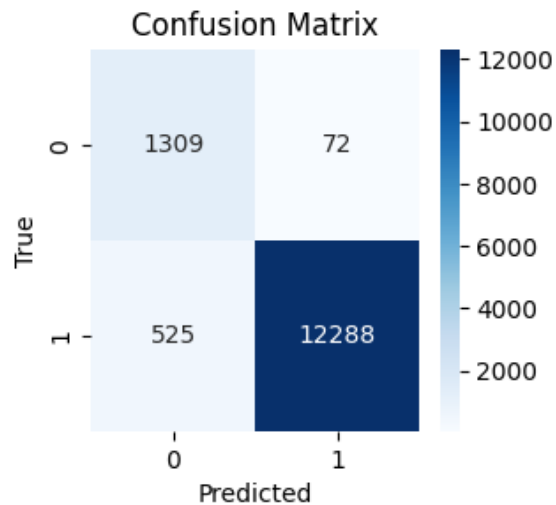
- Classes are heavily imbalanced (only 5 % are anomalies)
- For the anomaly case, rearrange to get double the amount of data.



- Balanced Cross-Entropy loss (basically weighted based on amount of imbalance)
- Optimizer = Adam, lr = 0.001, wd = 0.01

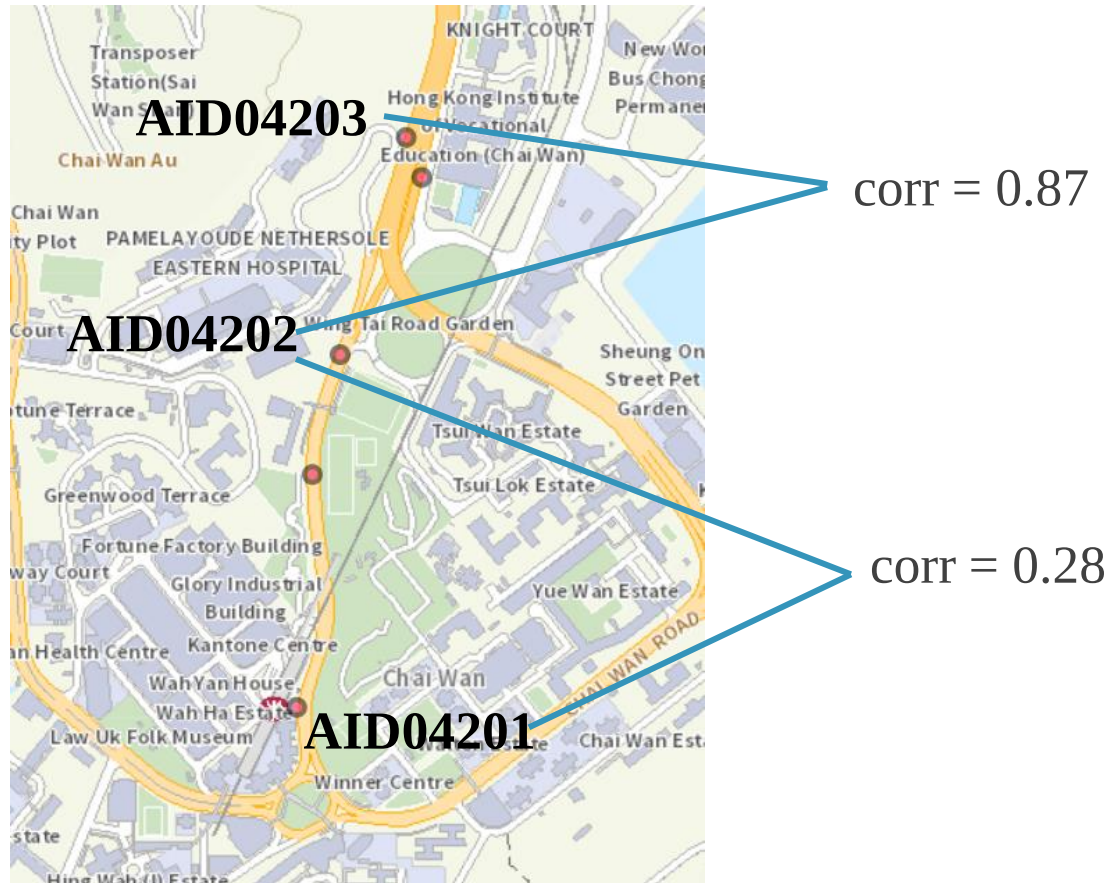
Training results of Supervised Model

- Balanced Accuracy for Test data = 96 %
- Note: Output is probability hence threshold used is <0.5 is anomaly.
- Recall for class '0' = 95%



Results

Correlation values



- Here we have calculated the correlation for entire 6 months
- Clearly AID04201 seems to be anomaly.
- Hence by analysing correlation within triplet we can identify evident anomalies.

Correlation values



AID04201



AID07102



AID08203

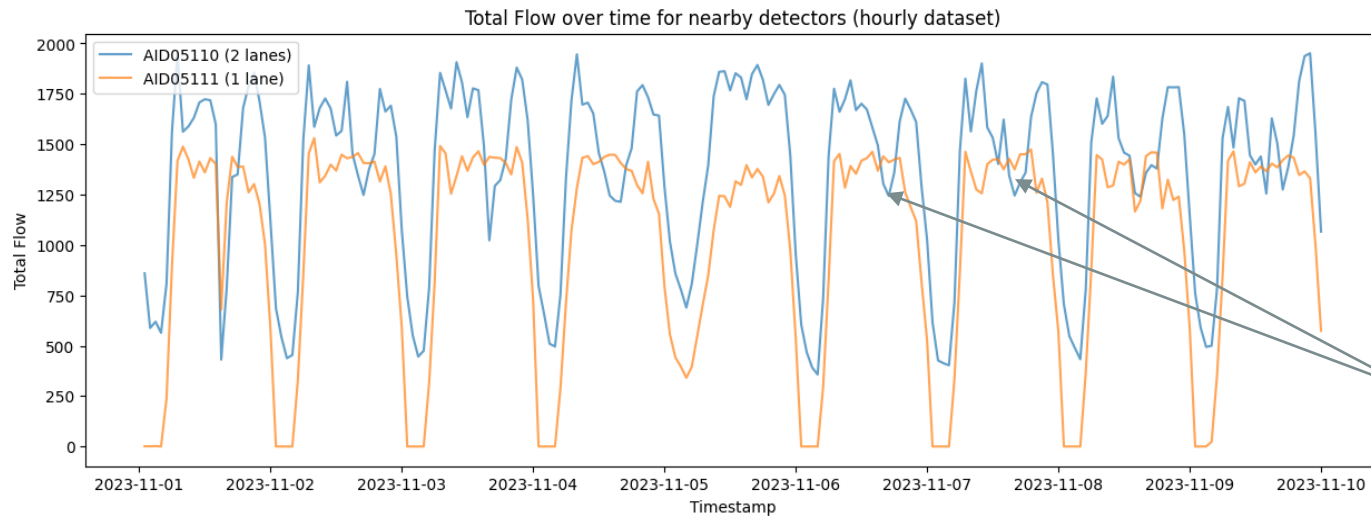
- These detectors consistently have extremely low correlation (< 0.4)

More examples of anomalies



AID05110 (westbound, 2 lanes) : Upstream detector of blue road.

AID05111 (westbound, 1 lane) : Downstream detector of yellow road.



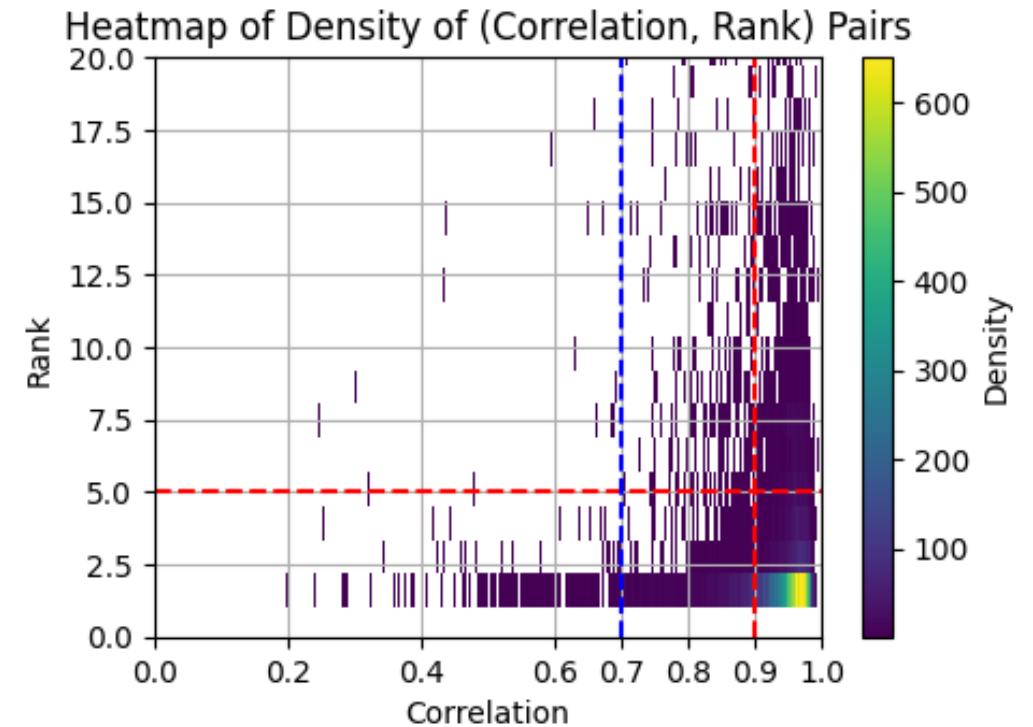
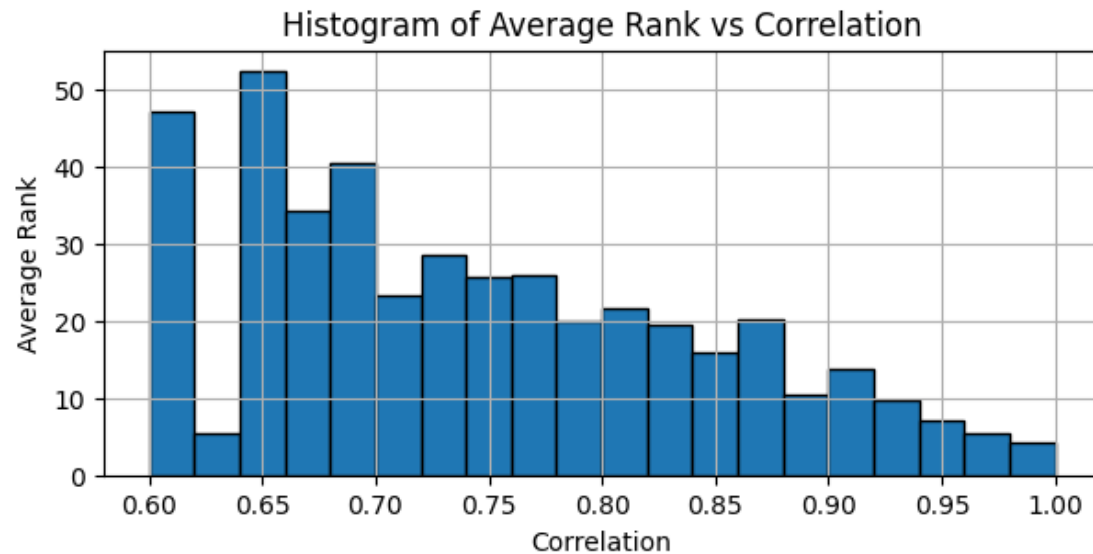
The blue road diverges into yellow and red road. We expect the total flow of blue to be greater than yellow at all times, but it gets violated in the graph.

Correlation = 0.78

Anomalies

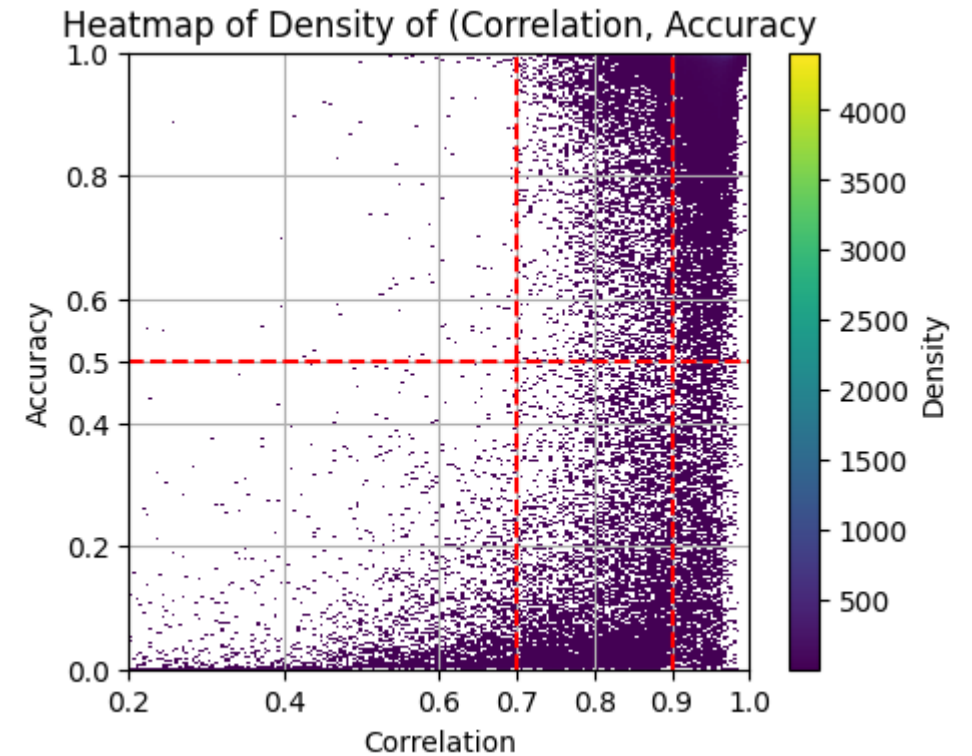
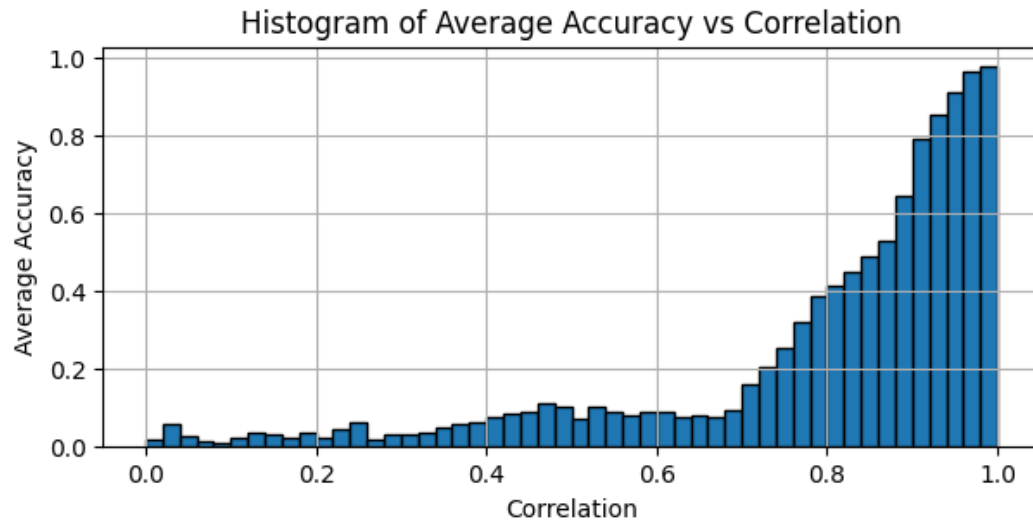
Profile Model vs Correlation

- For each day and detector we compare the profile model rank and the corresponding correlation.
- We can see that for higher correlation we are more likely to get better rank.



Supervised Model vs Correlation

- Similarly we compare the supervised model accuracy with correlation
- We can see that as expected higher correlation are more likely to have higher accuracy



Conclusion

Conclusion

- If the detector on a particular day has >0.9 correlation, we treat it normal, and if its <0.7 then anomaly.
- When the correlation is between 0.7 and 0.9, we take the help of profile model and the supervised model, and combining their results we take the decision.
- Using the 3 methods discussed we can automatically detect the anomalies among the network of detectors for that particular day.
- Possible future work includes lane-wise analysis, and to exploit speed-flow-occupancy relation.